Assignment 1 Solutions
NIS3352-01 Modern Cryptography (2)

Problem 1. Let $F_1, \ldots, F_p$ be events, where $p = \mathsf{poly}(\lambda)$, and suppose that for every $i \in [p]$,

$$\Pr[F_i] = 1 - \mathsf{negl}_i(\lambda)$$

for some negligible function negl_i. We show that

$$\Pr[F_1 \wedge F_2 \wedge \cdots \wedge F_p]$$

is overwhelming.

Let $\overline{F_i}$ denote the complement of F_i. Then

$$\Pr[\overline{F_i}] = 1 - \Pr[F_i] = \mathsf{negl}_i(\lambda).$$

By De Morgan's law,

$$\Pr\big[F_1 \wedge \cdots \wedge F_p\big] = 1 - \Pr\big[\overline{F_1} \vee \cdots \vee \overline{F_p}\big].$$

Applying the union bound,

$$\Pr\big[\overline{F_1} \vee \cdots \vee \overline{F_p}\big] \le \sum_{i=1}^{p} \Pr[\overline{F_i}] = \sum_{i=1}^{p} \mathsf{negl}_i(\lambda).$$

Since $p = \mathsf{poly}(\lambda)$, the sum of polynomially many negligible functions is still negligible. Hence there exists a negligible function $\mathsf{negl}'(\lambda)$ such that

$$\Pr\big[\overline{F_1} \vee \cdots \vee \overline{F_p}\big] \le \mathsf{negl}'(\lambda).$$

Therefore,

$$\Pr[F_1 \wedge \cdots \wedge F_p] \ge 1 - \mathsf{negl}'(\lambda),$$

which means that $\Pr[F_1 \wedge \cdots \wedge F_p]$ is overwhelming.

Problem 2. We prove that public-key encryption is IND-CPA secure if and only if it is IND-mCPA secure.

(IND-CPA $\Rightarrow$ IND-mCPA). Assume the scheme is IND-CPA secure. We show that it must also be IND-mCPA secure.

Suppose, for contradiction, that there exists an adversary $\mathcal{A}$ that breaks IND-mCPA with non-negligible advantage $\epsilon(\lambda)$. In the IND-mCPA game, $\mathcal{A}$ outputs two message vectors of equal length

$$\mathbf{M}_0 = (m_{0,1}, \ldots, m_{0,t}), \qquad \mathbf{M}_1 = (m_{1,1}, \ldots, m_{1,t}),$$

where $t = \mathsf{poly}(\lambda)$, and receives encryptions of either all messages in $\mathbf{M}_0$ or all messages in $\mathbf{M}_1$.

Define hybrids $H_0, H_1, \ldots, H_t$ as follows: in hybrid H_j, the challenger encrypts the first j messages from $\mathbf{M}_1$ and the remaining $t - j$ messages from $\mathbf{M}_0$. Thus,

- H_0 is the experiment in which all ciphertexts encrypt messages from $\mathbf{M}_0$;
- H_t is the experiment in which all ciphertexts encrypt messages from $\mathbf{M}_1$.

Since $\mathcal{A}$ distinguishes H_0 from H_t with advantage $\epsilon(\lambda)$, by a standard hybrid argument there exists some index $j \in \{0, \ldots, t-1\}$ such that $\mathcal{A}$ distinguishes H_j from H_{j+1} with advantage at least $\epsilon(\lambda)/t$, which is still non-negligible because t is polynomial.

Now construct an IND-CPA adversary $\mathcal{B}$ as follows. Given a public key pk, $\mathcal{B}$ forwards it to $\mathcal{A}$ and receives $\mathbf{M}_0, \mathbf{M}_1$. Then $\mathcal{B}$ submits the pair $(m_{0,j+1}, m_{1,j+1})$ to its IND-CPA challenger and receives a challenge ciphertext C^*. Next, $\mathcal{B}$ forms a vector of ciphertexts by encrypting the first j messages from $\mathbf{M}_1$, placing C^* in position $j+1$, and encrypting the remaining messages from $\mathbf{M}_0$. This perfectly simulates either H_j or H_{j+1}, depending on which message was encrypted in C^*. Finally, $\mathcal{B}$ outputs whatever bit $\mathcal{A}$ outputs.

Therefore, if $\mathcal{A}$ distinguishes H_j from H_{j+1} with non-negligible advantage, then $\mathcal{B}$ breaks IND-CPA with non-negligible advantage as well, contradicting IND-CPA security. Hence IND-CPA implies IND-mCPA.

(IND-mCPA $\Rightarrow$ IND-CPA). This direction is immediate. If a scheme is IND-mCPA secure, then it remains secure when the adversary chooses vectors of length 1. But that is exactly the IND-CPA game. Therefore IND-mCPA implies IND-CPA.

Combining both directions, IND-CPA and IND-mCPA are equivalent.

Problem 3. We show that the plain RSA encryption scheme is not IND-CPA secure.

The key observation is that plain RSA encryption is deterministic: for a fixed public key (N, e) and a message $M \in \mathbb{Z}_N^*$, the ciphertext is always

$$C = M^e \bmod N.$$

There is no randomness in the encryption algorithm.

We now describe an adversary $\mathcal{A}$ that wins the IND-CPA game with probability 1.

1. The challenger generates an RSA key pair and gives the public key (N, e) to $\mathcal{A}$.
2. $\mathcal{A}$ chooses two distinct messages $M_0, M_1 \in \mathbb{Z}_N^*$ and sends them to the challenger.
3. The challenger chooses a random bit $b \in \{0, 1\}$, computes
$$C^* = M_b^e \bmod N,$$
and sends C^* to $\mathcal{A}$.
4. Using the public key, $\mathcal{A}$ computes
$$C_0 = M_0^e \bmod N.$$
If $C^* = C_0$, then $\mathcal{A}$ outputs 0; otherwise it outputs 1.

Because RSA encryption is deterministic and $M_0 \neq M_1$, the ciphertexts of M_0 and M_1 are distinct. Hence $\mathcal{A}$ always recovers the correct bit b. Its success probability is therefore 1, so its advantage is also maximal.

Thus plain RSA encryption is not IND-CPA secure.

Problem 4. Consider the encryption scheme with public key (G, q, g, h) and private key x, where $h = g^x$. To encrypt a bit $b \in \{0, 1\}$:

- if $b = 0$, choose $y \xleftarrow{\$} \mathbb{Z}_q$ and output $(c_1, c_2) = (g^y, h^y)$;
- if $b = 1$, choose independent $y, z \xleftarrow{\$} \mathbb{Z}_q$ and output $(c_1, c_2) = (g^y, g^z)$.

Efficient decryption. Given the private key x and a ciphertext (c_1, c_2), compute c_1^x and compare it with c_2.

Indeed:

- If $b = 0$, then
$$c_1^x = (g^y)^x = g^{xy} = (g^x)^y = h^y = c_2.$$
So the equality always holds.
- If $b = 1$, then $c_2 = g^z$ for an independent random z, while
$$c_1^x = (g^y)^x = g^{xy}.$$
Since z is independent and uniform in $\mathbb{Z}_q$, the probability that $g^z = g^{xy}$ is exactly $1/q$, which is negligible in the security parameter.

Therefore the decryption algorithm is:

$$\text{output 0 if } c_2 = c_1^x, \quad \text{and output 1 otherwise.}$$

This is efficient and correct except with negligible error probability.

CPA security under the DDH assumption. We prove that if DDH is hard in G, then this encryption scheme is IND-CPA secure.

Suppose, toward contradiction, that there exists a PPT adversary $\mathcal{A}$ that distinguishes encryptions of 0 and 1 with non-negligible advantage. We build a PPT distinguisher $\mathcal{B}$ for DDH.

The DDH challenger gives $\mathcal{B}$ a tuple

$$(g, A, B, T) = (g, g^x, g^y, T),$$

where either $T = g^{xy}$ (the Diffie–Hellman case) or $T = g^z$ for uniform $z \xleftarrow{\$} \mathbb{Z}_q$ (the random case).

Algorithm $\mathcal{B}$ proceeds as follows:

1. Set $h := A = g^x$ and give the public key (G, q, g, h) to $\mathcal{A}$.
2. Give $\mathcal{A}$ the challenge ciphertext
$$(c_1, c_2) := (B, T).$$
3. Let b' be the bit output by $\mathcal{A}$. Output "DDH" if $b' = 0$, and output "random" if $b' = 1$.

Now analyze the simulation:

- If $T = g^{xy}$, then
$$(c_1, c_2) = (g^y, g^{xy}) = (g^y, h^y),$$
which is distributed exactly as an encryption of 0.

- If $T = g^z$ for uniform z, then

  $$(c_1, c_2) = (g^y, g^z),$$

  which is distributed exactly as an encryption of 1.

Thus any non-negligible advantage of $\mathcal{A}$ in distinguishing encryptions of 0 and 1 translates directly into a non-negligible advantage for $\mathcal{B}$ in solving DDH, contradicting the DDH assumption.

Therefore the scheme is IND-CPA secure under the DDH assumption.